

Algorithm Analysis

Start Assignment

- Due Sunday by 11:59pm
- Points 10
- Submitting a file upload
- Available Sep 6 at 12am - Sep 16 at 11:59pm

Write a program to verify that the `ArrayList` `indexOf` method is $O(n)$; call the program `AlgorithmAnalysis.java`

Then, write a document (either Word, LibreOffice, PDF, or plain text) that describes and summarizes the results you obtained.

This is a very open-ended assignment. You can decide how to do the verification, the number of items to put in the `ArrayList`, the number of times you want to call `indexOf`, etc. This is all entirely up to you, but you must give that information in your summary document.

The summary document must contain your name and the date when it was written.

[This file gives an example of a summary of a student's program.](#)

<https://sjeccd.instructure.com/courses/53429/files/12442867?wrap=1> **You do not have to use the exact same setup for your program.** If you feel that a different approach would be more effective, use it—just make sure you document it!

Since this is not an English class, I will not take off for spelling, grammar, or punctuation errors unless there are so many that it makes the document extremely difficult to understand. (I will definitely appreciate it if you proofread your document!)

When You Finish

- Remember that your program should be named `AlgorithmAnalysis.java`
- Name your summary document in the form `lastname_firstname_algorithm.ext`, where `ext` is the appropriate extension for your file type (docx, pdf, etc.).
- Upload both files to Canvas

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Criteria	Ratings			Pts
Programming Style	2 to >1.0 pts Good	1 to >0.0 pts Fair	0 pts Poor	2 pts
Program matches description How well does the program match the description given in the summary document?	2 to >1.0 pts Matches well	1 to >0.0 pts Matches somewhat	0 pts Does not match	2 pts
Description of tests Report describes in detail the steps used to verify the hypothesis.	2 to >1.0 pts Excellent	1 to >0.0 pts Fair	0 pts Poor	2 pts
Summary Statistics in report Document provides the appropriate statistics and analysis to prove linear relationship	2 to >1.0 pts Excellent	1 to >0.0 pts Fair	0 pts Poor	2 pts
Readability of Document Is the summary document understandable?	2 to >1.0 pts Very Clear	1 to >0.0 pts Somewhat Clear	0 pts Not Clear	2 pts
Total Points: 10				